

Q1 - More units and waves: Let's say we have a fat ribbon instead of a string vibrating. Instead of a mass per unit length, the ribbon has a mass per unit area σ . Instead of a tension, we now have a tension per unit length τ (if we have the same total force on a wider ribbon, the force on each bit will drop, and the wave speed will go down). What are the units of mass per area σ and force per length τ in kg, m, and s? What would the (approximate?) wave speed in the ribbon be in terms of σ and τ ?

The units of σ are mass per unit area. Area is in m^2 , mass is in kg , and so σ has units of kg/m^2 . The units of τ are force per unit length or newtons per metre. That is $kgm/s^2/m = kg/s^2$. To get a speed, at the very least we need to cancel out the kg. Both σ and τ have a kg on top, so we can divide them by each other to get rid of kg . If we take σ/τ , we get

$$\frac{kg/m^2}{kg/s^2} = \frac{s^2}{m^2} \quad (1)$$

We could get to a speed from here, but we're better off taking τ/σ :

$$\tau/\sigma = \frac{kg/s^2}{kg/m^2} = \frac{m^2}{s^2} \quad (2)$$

We want m/s , we have m^2/s^2 , so we can just take a square root. That gives a speed of $\sqrt{\tau/\sigma}$. Sanity check - if we increase the tension τ , we expect the speed to go up. That works. If we increase the mass, the speed should go down. That also checks out, so our answer passes sanity tests.

Q2 - Repeating Middle C: What is the frequency of middle C (C4)? A4 (which is higher) has a frequency of 440 Hz.

Rather than jumping down to A3 and counting up, let's count down from A4 to get to C4. We have to go through G#, G, F#, F, E, D#, D, and C# to get to C. If you count, that's 9 half-steps. That means C4 should have the frequency of A4 divided by r^9 where as usual r is the twelfth root of 2 $2^{1/12}$. Writing out, we get the frequency is $440/r^9 = 440/2^{9/12} = 440/2^{3/4} = 261.6Hz$, as expected.

Q3 - An ideal simple flute (with no holes) would have a lowest note where the wavelength is four times longer than the flute. If the lowest note of a flute is a middle C (261 Hz), how long should the flute be? The wave speed in the flute is the speed of sound. Your professor has finally realized that a real flute is not ideal because it has a big hole at the end where the flutist is blowing. In this case, the flute is open at both ends, and behaves like a string nailed at both ends. How long should the flute be now? Does this line up with the true value (about 610 mm, according to the web)?

The velocity of a wave is the wavelength times the frequency $v = \lambda\nu$. If you're fuzzy on this, remember that the frequency ν is the number of full waves that pass us every second, and the length of each full wave is the wavelength λ . If say I have 20 waves per second, and each wave is 5m long, then if I need to get 100m worth of waves passing me every second. We'll take the speed

of sound at a comfortable temperature to be $340m/s$ and we know that the frequency of middle C is 261 Hz. If we plug the values for v and ν into $v = \lambda\nu$, we have

$$340m/s = \lambda 261/s \quad (3)$$

or $\lambda = 340/261m = 1.30m$. The length of an ideal flute that is closed at one end and open at the other is a quarter of a wavelength, or $0.32m$ or $32cm$. That's about half the length of an actual flute. A flute that is open at both ends or closed at both ends has length $\lambda/2 = 0.65m$. That's much closer to the actual length of a flute (about $60cm$).

Q4 - We saw that the wave speed on a violin's E-string is about 250 m/s. The piano's highest note is a C8 - 4 octaves above middle C. What frequency is this note? If the wave speed is about the same on a high piano string and a high violin string, how long would the piano string be? In practice, the piano strings are under a lot more tension and so the highest notes have a wave speed about double that of a violin.

We know that middle C/C4 is just over 261 Hz. We are 4 octaves above this. Each octave gives a factor of 2, so the pitch is a factor of $2 * 2 * 2 * 2 = 16$ times higher. The frequency is then $261 \times 16 = 4186Hz$. If we have a wave speed of $250m/s$, then again using $v = \nu\lambda$, we have $250 = 4186\lambda$ or $\lambda = 250/4186 = 0.06m = 6cm$. If we're going at the fundamental, the length of the string would be half that, or $3cm$. As mentioned, the piano strings are under a lot more tension, so the real answer is that the wave speed is closer to $500m/s$.

Q5 - A piano has 88 keys. What is the ratio of the fundamental frequencies of the highest and lowest keys? If the vibrating length of the highest pitched string is 65mm, how long would the bottom string have to be if made of same wire under the same tension?

If there are 88 keys, we have to go down by 87 half-steps to get from the top to the bottom. That means the ratio of frequencies is going to be $r^{87} = 2^{87/12} = 152.2$. If the strings are the same weight per length and under the same tension, then the bottom string would be 152 times longer than the top string. Since the top string is 6.5 cm, the bottom string would need to be 10m long. Nobody could afford that! The bass strings are much thicker in order to slow the wave speed down while still maintaining tension (you need tension in order for the vibrating string to push hard enough on the sound board for it to vibrate. Those are the vibrations that actually push the air around that you hear). Incidentally, I strongly encourage you to look up the Alexander Piano - the longest piano in the world.